

# Assignment III : Ex-Post ML techniques for Portfolio Optimization

## Data:

Fetch the adjusted close price data for the last 5 years of these 12 stocks from NSE using yfinance.

Use Monthly returns for Part 1 of the assignment and Daily returns for Part 2 of the Assignment.

| Tickers for yfinance                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| TCS.NS, INFY.NS, HCLTECH.NS, TATASTEEL.NS, HINDALCO.NS, JINDALSTEL.NS, HDFCBANK.NS, ICICIBANK.NS, KOTAKBANK.NS, HINDUNILVR.NS, ITC.NS, NESTLEIND.NS |

Data Partitioning: Train your model using the first 4 years of the dataset. Reserve the final year (Year 5) exclusively for testing to evaluate the model's predictive performance.

## Part 1: Statistical Regularization via Ledoit-Wolf Shrinkage

Standard Markowitz optimization is highly sensitive to estimation errors in the covariance matrix. In this section, you will implement shrinkage methods to improve the stability of the efficient frontier. You will verify this sensitivity using bootstrap resampling to demonstrate how significantly the sample covariance matrix fluctuates across different data draws.

### Task 1.1: The Ill-Conditioned Matrix

Calculate the Sample Covariance Matrix ( $S$ ) for your 12-stock NSE universe using the provided 4-year returns. Compute the 'Condition Number' ( $\kappa$ ) of the matrix to demonstrate that it is ill-conditioned.

### Task 1.2: Shrinkage Implementation

Apply the Ledoit-Wolf Shrinkage estimator to compute a regularized covariance matrix. Compare the resulting 'Maximum Sharpe' portfolio weights against those generated using the non-regularized sample covariance.

## Part 2: Structural Regularization via Hierarchical Risk Parity (HRP)

Hierarchical Risk Parity (HRP) uses Machine Learning (Clustering) to regularize the portfolio by addressing the underlying topology of risk, rather than just shrinking the numerical entries of the covariance matrix. The study includes cluster discovery along with recursive bisection.

Compare the Portfolio Turnover (total change in weights) between HRP and a standard MPT model.

### Deliverables:

**Python Code:** Implementation of Ledoit-Wolf shrinkage and the HRP recursive bisection algorithm.

#### 2 Pager Report with

- Visualization 1: Side-by-side Heatmap (Sample vs. Shrunk Covariance).
- Visualization 2: HRP Dendrogram with labeled statistical clusters.
- Visualization 3: Weight Sensitivity Plot (Comparing MPT vs. Regularized weights during the Banking Crisis).
- Reflection : A summary on why regularization is a prerequisite for real-world quantitative trading.

### Grading Rubric

| Criteria                    | Weight | Expectation                                                        |
|-----------------------------|--------|--------------------------------------------------------------------|
| Mathematical Implementation | 40%    | Correct derivation of Shrinkage Intensity and HRP bisection logic. |

|                          |     |                                                                               |
|--------------------------|-----|-------------------------------------------------------------------------------|
| Stability Analysis       | 20% | Demonstrating reduced turnover in regularized models.                         |
| Cluster Interpretation   | 20% | Insightful labeling of the dendrogram without relying on pre-defined sectors. |
| Results and Presentation | 20% | -                                                                             |